PROPERLY PROXIMAL GRAPH PRODUCT VON NEUMANN ALGEBRAS

1. Introduction

The notion of proper proximality for groups was introduced in [BIP21], generalizing Ozawa's notion of biexactness from [Oza04]. This notion is a stengthening of non–inneramenability in the sense of Effros, and has applications to the study of von Neumann algebras. In particular, it is an invariant upto measure equivalence and von Neumann equivalence [IPR19], and when combined with W*CBAP it yields absence of Cartan subalgbras, following ideas of [OP10a].

In the paper [DKE24], proper proximality was classified for graph products of groups, in the sense of Green [Gre90]. In this short note, this is generalized to the setting of graph products of tracial von Neumann algebras [CF17]. Indeed the notion of proper proximality has been extended to von Neumann algebras in [DEP23], so this question is natural. First we prove the following result:

Theorem 1.1. *Let Γ be a finite graph with $|\mathcal{V}(\mathcal{G})| \geq 3$, and (M_v, φ_v) be separable tracial for each $v \in \mathcal{V}$. If Γ is irreducible (in the sense that there does not exists subgraphs $\mathcal{G}_1, \mathcal{G}_2$ such that $\mathcal{V}(\mathcal{G}_1) \cup \mathcal{V}(\mathcal{G}_2) = \mathcal{V}(\mathcal{G})$ and every $(v,u) \in \mathcal{E}(\mathcal{G})$ for all $v \in \mathcal{V}(\mathcal{G}_1), u \in \mathcal{V}(\mathcal{G}_2)$), and each M_v has a trace zero unitary, then M is properly proximal.*

A full classification for graph products then follows by combining this statement with the known fact from [DEP23] that a tensor product of finite von Neumann algebras $N_1 \overline{\otimes} N_2$ is properly proximal if and only if both are. Indeed, every graph can be decomposed into combinations of the above and irreducible subgraphs.

For the sake of brevity, we will quickly recall and follow the standard notations from [DEP23], and assume that the reader is familiar with standard notions and arguments in von Neumann algebra theory.

2. Proof of main result

Let $(\mathcal{G}, \mathcal{E})$ be a finite graph, let $\{(M_v, \varphi_v) : v \in \mathcal{V}\}$ be a family of von Neumann algebras with faithful normal states. Consider the graph product von Neumann algebra (M, φ).

Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be two subgraphs, and M_1, M_2 the corresponding graph product von Neumann algebras, considered as subalgebras of M. Let also $N = M_1 \cap M_2$ be the graph product of the subgraph $\mathcal{G}_1 \cap \mathcal{G}_2$.

Let e_i, $i = 1, 2$ be the projection onto the subspace $L^2(M_i) \subset L^2(M)$. And e be the projection onto $L^2 N$.

Consider the the M-boundary pieces $\mathbb{X}_{M_i} = \overline{MJMJe_i\mathbb{B}(L^2M)e_iJMJM}^{\|\cdot\|}$, $\mathbb{X}_N = \overline{MJMJe\mathbb{B}(L^2M)eJMJM}^{\|\cdot\|}$, and their M-M JMJ-JMJ closure $\mathbb{K}^{\infty,1}_{M_i}(M)$, $\mathbb{K}^{\infty,1}_{N}(M)$.

Let $\iota^\sharp : \mathbb{B}(L^2M) \to \mathbb{B}(L^2M)^{\sharp *}_J$ be the canonical embedding.

Let $q_i = \vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJe_iJv^*Ju^*)$ be the identity of $(\mathbb{X}_{M_i})^{\sharp *}_J \subset \mathbb{B}(L^2M)^{\sharp *}_J$ and $q = \vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJeJv^*Ju^*)$ be the identity of $(\mathbb{X}_N)^{\sharp *}_J$.

We want to show that $\mathbb{K}^{1,\infty}_{M_1}(M)\cap\mathbb{K}^{1,\infty}_{M_2}(M) = \mathbb{K}^{1,\infty}_{N}(M)$. But notice that $\mathbb{K}^{1,\infty}_{M_i}(M) = (\iota^\sharp)^{-1}\left(\mathbb{X}^{\sharp *}_{M_i}\right)$, it suffices to check that

$$(\mathbb{X}_{M_1})^{\sharp *}_J \cap (\mathbb{X}_{M_2})^{\sharp *}_J = (\mathbb{X}_N)^{\sharp *}_J.$$

Finally, since $(\mathbb{X}_{M_i})^{\sharp *}_J = q_i\mathbb{B}(L^2M)^{\sharp *}_J q_i$, this amounts to show that $q_1q_2 = q_2q_1 = q$.

Lemma 2.1. *Let $\mathcal{K}$ be a normal representation of $\mathbb{B}(L^2M)^{\sharp *}_J$ so that $\mathbb{B}(L^2M)^{\sharp *}_J \subset \mathbb{B}(\mathcal{K})$. Then $\iota^\sharp(e_1)q_2\mathcal{K} = \overline{\iota^\sharp(e_1MJMJe_2)\mathcal{K}} \subset \overline{\iota^\sharp(MJMJe)\mathcal{K}} = q\mathcal{K}$.*

Proof. Note that

$$\begin{aligned} e_1MJMJe_2 =& e_1(M\ominus M_1)J(M\ominus M_1)Je_2 + e_1M_2J(M\ominus M_1)Je_2 + e_1MJM_1Je_2 \\ =& e_1(M\ominus M_1)J(M\ominus M_1)Je_2 + M_1e_1J(M\ominus M_1)Je_2 \\ &+ JM_1Je_1(M\ominus M_1)e_2 + JM_1JM_1e. \end{aligned}$$

So, it suffices to show that the range of those operators belongs to $\iota^\sharp(MJMJe)\mathcal{K}$.

For this, we first claim that $(e_1-e)(M\ominus M_1)J(M\ominus M_1)Je_2 = 0$, or equivalently $e_2(M\ominus M_1)J(M\ominus M_1)J(e_1-e) = 0$. Indeed, a vector in $(M\ominus M_1)J(M\ominus M_1)J(e_1-e)L^2M$ is spanned by linear combinations of vectors in $\mathcal{H}_{v_1\cdots v_n}$ with $v_1\cdots v_n$ a reduced word in $\mathcal{V}(\mathcal{G})$ such that $v_j\notin\mathcal{V}(\mathcal{G}_2)$ for sum j, but this guarantees that the vector is orthogonal to L^2M_2.

Similarly, we have $(e_1-e)J(M\ominus M_1)Je_2 = (e_1-e)(M\ominus M_1)e_2 = 0$. Therefore, we have $\iota^\sharp(e_1(M\ominus M_1)J(M\ominus M_1)Je_2)\mathcal{K} \subset \iota^\sharp(e(M\ominus M_1)J(M\ominus M_1)Je_2)\mathcal{K} \subset \iota^\sharp(MJMJe)\mathcal{K}$. Similarly, we have $M_1e_1J(M\ominus M_1)Je_2$, $JM_1Je_1(M\ominus M_1)e_2 \subset \iota^\sharp(MJMJe)\mathcal{K}$. Combining those inclusions, we obtain $\iota^\sharp(e_1MJMJe_2)\mathcal{K} \subset \iota^\sharp(MJMJe)\mathcal{K}$. □

Corollary 2.2. *$[q_1,q_2]=0$ and $q_1q_2 = q$.*

Proof. Since q_2 commutes with $\iota^\sharp(M)$ and $\iota^\sharp(JMJ)$, and $q_1 = \vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJe_1Jv^*Ju^*)$, to show $[q_1,q_2]=0$, it suffices to check that $[\iota^\sharp(e_1),q_2]=0$. But by the above lemma, we have $\iota^\sharp(e_1)q_2 = q\iota^\sharp(e_1)q_2 = q_2\iota^\sharp(e_1)q_2$. Taking the adjoint, we obtain $\iota^\sharp(e_1)q_2 = q_2\iota^\sharp(e_1)q_2 = \iota^\sharp(e_1)q_2$.

Note that as $[\iota^\sharp(e_1),q_2]=0$, we have

$$q_1q_2 = q_1 = \vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJe_1Jv^*Ju^*)q_2 = \vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJ)\iota^\sharp(e_1)q_2\iota^\sharp(Jv^*Ju^*).$$

Since $\iota^\sharp(e_1)q_2 = q\iota^\sharp(e_1)q_2 = q_2\iota^\sharp(e_1) = q_2\iota^\sharp(e_1)q = q\iota^\sharp(e_1)q$, we obtain

$$\begin{aligned} q_1q_2 =& \vee_{u,v\in\mathcal{U}(M)}\ \iota^\sharp(uJvJ)q\iota^\sharp(e_1)q\iota^\sharp(Jv^*Ju^*) \\ =& q\left(\vee_{u,v\in\mathcal{U}(M)}\iota^\sharp(uJvJe_1Jv^*Ju^*)\right)q = qq_1q = q. \end{aligned}$$

□

Therefore we immediately obtain the following:

Corollary 2.3. $\mathbb{K}^{1,\infty}_{M_1}(M) \cap \mathbb{K}^{1,\infty}_{M_2}(M) = \mathbb{K}^{1,\infty}_{N}(M)$. □

Lemma 2.4. *Suppose that for is $v \in \mathcal{V}(\mathcal{G})$ (M_v, φ_v) is separable and tracial. If $\{\mathcal{G}_i\}_{i=1,\cdots,s}$ is a family of subgraphs of $\mathcal{G}$ and M is properly proximal relative to $\Gamma(\mathcal{G}_i)$ for each i, then M is properly proximal relative to $\Gamma(\cap_i \mathcal{G}_i)$.*

Proof. Suppose that M is not properly proximal relative to $\Gamma(\cap_i \mathcal{G}_i)$, then by [DEP23, Lemma 8.5] there is a M-central state φ on $\tilde{\mathbb{S}}_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}}(M)$ such that $\varphi|_M$ is normal.

Let $q_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}}$ be the identity of $(\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)})^{\sharp *}_J$. If $\varphi(q_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}}) \neq 0$, then $\frac{1}{\varphi(q_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}})} \varphi \circ \mathrm{Ad}(q_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}})$ is a M-central state φ on $\tilde{\mathbb{S}}_{\mathbb{X}_{\mathcal{G}_i}}(M)$ for each i (as $\mathrm{Ad}(q_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}})\mathbb{B}(L^2M)^{\sharp *}_J \subset (\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)})^{\sharp *}_J \subset (\mathbb{X}_{\Gamma(\mathcal{G}_i)})^{\sharp *}_J \subset \tilde{\mathbb{S}}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}}(M)$), which is a contradiction to the fact that M is properly proximal relative to $\Gamma(\mathcal{G}_i)$.

If $\varphi(q^{\perp}_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}}) = 1$, then as $q^{\perp}_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}} = \vee_i q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}}$ (by Corollary 2.3), there must be a i such that $\varphi(q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}}) \neq 0$. Then, since $\mathrm{Ad}(q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}})(\tilde{\mathbb{S}}_{\mathbb{X}_{\Gamma(\cap_i \mathcal{G}_i)}}(M)) \subset q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}} \mathbb{B}(L^2M)^{\sharp *}_J q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}} \cap \iota^{\sharp}(JMJ)' \subset \tilde{\mathbb{S}}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}}(M)$, we have that $\frac{1}{\varphi(q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}})}\varphi \circ \mathrm{Ad}(q^{\perp}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}})$ is a M-central state φ on $\tilde{\mathbb{S}}_{\mathbb{X}_{\Gamma(\mathcal{G}_i)}}(M)$, again contradicting to the fact that M is properly proximal relative to $\Gamma(\mathcal{G}_i)$. □

Lemma 2.5. *Let M_1, M_2 be two separable tracial von Neumann algebras with a common expected subalgebra B. Suppose that there exists unitaries $u_1, u_2 \in \mathcal{U}(M_1)$ and $u_3 \in \mathcal{U}(M_2)$ with $E_B(u_1) = E_B(u_2) = E_B(u_3) = E_B(u_1^* u_2) = 0$, then the amalgamated free product $M := M_1 *_B M_2$ is properly proximal relative to B.*

Proof. First, we recall the notation $\lambda_i(e_B)$ is defined as the projection onto the space of vectors in L^2M beginning with $L^2M^o_{i+1}$ or L^2B. Also, we have $\lambda_i(e_B) \in \mathbb{S}_B(M)$ [TY25, Prop. 4.16].

Suppose that M is not properly proximal relative to B, then there exists a M-central state φ on $\tilde{\mathbb{S}}_B(M)$ that is normal on M. We first claim that $\varphi(\iota^{\sharp}(e_B)) = 0$. Indeed, suppose $\varphi(\iota^{\sharp}(e_B)) \neq 0$, then $\varphi \circ \mathrm{Ad}(\vee_{u \in \mathcal{U}(M)} \iota^{\sharp}(u e_B u^*))$ is a M-central positive functional on $\mathbb{B}(L^2M)$ not vanishing on $M e_B M$ (this is because the projections $\vee_{u \in \mathcal{U}(M)} \iota^{\sharp}(u e_B u^*)$ commutes with every unitary in M). By [OP10b, Lemma 3.3] (or [BH18, Theorem 2]), this implies that $M \preceq_M B$ which is impossible (as $\|E_B(x(u_1u_2)^n y)\|_2 \xrightarrow{n\to\infty} 0$ for all $x, y \in M$). Now, notice that $\lambda_1(e_B) + \lambda_2(e_B) = 1 + e_B$, so $\varphi(\iota^{\sharp}(\lambda_1(e_B) + \lambda_2(e_B))) = 1$. But since

$$u_1^* \lambda_1(e_B) u_1 + u_2^* \lambda_1(e_B) u_2 \leq \lambda_2(e_B), \quad u_3^* \lambda_2(e_B) u_3 \leq \lambda_1(e_B),$$

we obtain $2\varphi(\iota^{\sharp}(\lambda_1(e_B))) \leq \varphi(\iota^{\sharp}(\lambda_2(e_B))) \leq \varphi(\iota^{\sharp}(\lambda_1(e_B)))$ as φ is M-central. This implies that $\varphi(\iota^{\sharp}(\lambda_1(e_B))) = \varphi(\iota^{\sharp}(\lambda_2(e_B))) = 0$, contradicting to the fact that $\varphi(\iota^{\sharp}(\lambda_1(e_B) + \lambda_2(e_B))) = 1$. □

Theorem 2.6. *Let Γ be a finite graph with $|\mathcal{V}(\mathcal{G})| \geq 3$, and (M_v, φ_v) be separable tracial for each $v \in \mathcal{V}$. If Γ is irreducible (in the sense that there does not exists subgraphs $\mathcal{G}_1, \mathcal{G}_2$ such that $\mathcal{V}(\mathcal{G}_1) \cup \mathcal{V}(\mathcal{G}_2) = \mathcal{V}(\mathcal{G})$ and every*

$(v,u) \in \mathcal{E}(\mathcal{G})$ for all $v \in \mathcal{V}(\mathcal{G}_1), u \in \mathcal{V}(\mathcal{G}_2)$), and each M_v has a trace zero unitary, then M is properly proximal.

Proof. For each $v \in \mathcal{V}(\mathcal{G})$, we have

$$M = \Gamma(\mathcal{G}/\{v\}) *_{\Gamma(\mathrm{link}(v))} \Gamma(\mathrm{star}(v)).$$

Let $M_1 = \Gamma(\mathcal{G}/\{v\})$, $B = \Gamma(\mathrm{link}(v))$, and $M_2 = \Gamma(\mathrm{star}(v))$. Since Γ is irreducible, either M_1 must contains two unitaries u_1, u_2 such that $E_B(u_1) = E_B(u_2) = E_B(u_1^* u_2) = 0$ (otherwise we must have that there is only one vertex u in $\mathcal{V}(\mathcal{G})/\mathcal{V}(\mathrm{star}(v))$. Now, we must have $(u, u') \in \mathcal{V}(\mathcal{G})$, as otherwise free independency of M_u and $M_{u'}$ would provides the desired unitaries. But this implies that $\mathcal{G}$ can be decomposed into $\{u, v\}$ and $\mathcal{V}(\mathcal{G})/\{u, v\}$, contradicting to that $\mathcal{G}$ is irreducible). Therefore, by the above Lemma, M is properly proximal relative to $\Gamma(\mathrm{link}(v))$ for each v, and thus properly proximal relative to $\Gamma(\cap_{v\in\mathcal{V}(\mathcal{G})}\mathrm{link}(v)) = \mathbb{C}$. □

References

[BH18] Rémi Boutonnet and Cyril Houdayer. Amenable absorption in amalgamated free product von Neumann algebras. *Kyoto J. Math.*, 58(3):583–593, 2018.

[BIP21] Rémi Boutonnet, Adrian Ioana, and Jesse Peterson. Properly proximal groups and their von Neumann algebras. *Ann. Sci. Éc. Norm. Supér. (4)*, 54(2):445–482, 2021.

[CF17] Martijn Caspers and Pierre Fima. Graph products of operator algebras. *J. Noncommut. Geom*, 11(1):367–411, 2017.

[DEP23] Changying Ding, Srivatsav Kunnawalkam Elayavalli, and Jesse Peterson. Properly proximal von Neumann algebras. *Duke Math. J.*, 172(15):2821–2894, 2023.

[DKE24] Changying Ding and Srivatsav Kunnawalkam Elayavalli. Proper proximality among various families of groups. *Groups Geom. Dyn.*, 18(3):921–938, 2024.

[Gre90] Elisabeth Ruth Green. *Graph products of groups*. PhD thesis, University of Leeds, 1990.

[IPR19] Ishan Ishan, Jesse Peterson, and Lauren Ruth. Von Neumann equivalence and properly proximal groups. arXiv:1910.08682, 2019.

[OP10a] Narutaka Ozawa and Sorin Popa. On a class of II_1 factors with at most one Cartan subalgebra. *Ann. of Math. (2)*, 172(1):713–749, 2010.

[OP10b] Narutaka Ozawa and Sorin Popa. On a class of II_1 factors with at most one Cartan subalgebra, II. *Amer. J. Math.*, 132(3):841–866, 2010.

[Oza04] Narutaka Ozawa. Solid von Neumann algebras. *Acta Math.*, 192(1):111–117, 2004.

[TY25] Kai Toyosawa and Zhiyuan Yang. On relative biexactness of amalgamated free product von neumann algebras. *arXiv preprint arXiv:2505.19508*, 2025.